7. $\frac{1}{6}\log\left|\frac{e^x-3}{e^x+3}\right|+c$ 8. $\frac{1}{3}\log\left|3x+\sqrt{9x^2-7}\right|+c$ 9. $\log\left|(x+3)+\sqrt{x^2+6x+13}\right|+c$

10. $\log\left|\left(x-\frac{3}{2}\right)+\sqrt{x^2-3x+2}\right|+c$ 11. $\frac{1}{4}\log\left|x^4+\sqrt{x^8-1}\right|+c$

12. $\frac{\left(x+\frac{1}{2}\right)}{2}\sqrt{1+x+x^2}+\frac{3}{8}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{1+x+x^2}\right|+c$

13. $\frac{x}{2}\sqrt{x^2-2}-\log\left|x+\sqrt{x^2-2}\right|+c$ 14. $\frac{1}{4}\left[2x\sqrt{4x^2-5}-5\log\left|2x+\sqrt{4x^2-5}\right|\right]+c$

15. $\left(\frac{x+1}{2}\right)\sqrt{2x^2+4x+1}-\frac{\sqrt{2}}{4}\log\left|\sqrt{2}(x+1)+\sqrt{2x^2+4x+1}\right|+c$

16. $\frac{x^2}{2}-\frac{x}{2}\sqrt{x^2-1}+\frac{1}{2}\log\left|x+\sqrt{x^2-1}\right|+c$

## பயிற்சி 2.8

I:1. $\frac{1}{2}\left[e^2-1\right]$ 2. $\frac{1}{6}$ 3. $\frac{1}{2}\log\left[\frac{5}{2}\right]$ 4. $\log\left[\frac{1+e^3}{2}\right]$

5. $\frac{1}{2}\left[e-1\right]$ 6. $\frac{3}{8}$ 7. $\log\left[\frac{11}{5}\right]$ 8. 2 9. $\frac{1}{2}\left[2\log 2-1\right]$

II:1. 37 2. 4 3. 1 4. $c=4$

## பயிற்சி 2.9

1. 0 2. $\frac{\pi}{2}$ 3. 0 4. $\frac{\pi}{4}$ 5. 0 6. $\frac{16}{5}$

## பயிற்சி 2.10

1.( i) 6 (ii) $\frac{105\sqrt{\pi}}{16}$ (iii) $\frac{6!}{m^7}$ (iv) $\frac{3}{128}$ (v) $\left(2^6\right)5!$ 2. $\frac{1}{4}$

## பயிற்சி 2.11

1. $\frac{9}{2}$ 2. 4 3. 14 4. $\frac{1}{3}$

## பயிற்சி 2.12

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (b) | (c) | (a) | (a) | (a) | (b) | (b) | (a) | (d) | (c) | (b) | (b) | (b) | (b) | (c) |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| (d) | (c) | (c) | (b) | (a) | (b) | (b) | (c) | (a) | (a) | (a) | (b) | (d) | (b) | (c) |



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